Power & USB

File: power_usb.kicad_sch

ESP32

File: mcu.kicad_sch

H1
M3

H2
M3

H3
M3

H4
M3

HV supply

File: hv_supply.kicad_sch

Display

File: display.kicad_sch

electroNIX–USB: USB–C powered ESP32 nixie clock, 4 x Z573M
(one generator, 4– or 6–tube build: python3 source/gen.py OUTDIR ––tubes 6)

Electrical design follows TESTA–QUADRA electroNIXclock 3 (PCB–036, 2013),
the micro–USB powered board: 5 V boost + 7–diode multiplier to 170 V,
MMBTA42/MMBTA92 discrete tube drivers, neon colon, light sensor.

Changes from PCB–036:
- ATmega16 replaced by ESP32–WROOM–32E; runs stock firmware unchanged
  (4 tubes: BOARD_ELECTRONIX_3, 6 tubes: BOARD_ELECTRONIX_4_6T)
- USB–C (5k1 sink) + polyfuse + TVS + USBLC6, CH340C for programming
- no buzzer, no RTC crystal / supercap: time from NTP
- HV feedback clamp to 3V3, base resistors resized for 3.3 V drive
- two independent colon neons (SEC_0, SEC_1)

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ESP32 rework of TESTA–QUADRA electroNIXclock 3 (PCB–036), USB powered

**electroNIX–USB (Z573M)**

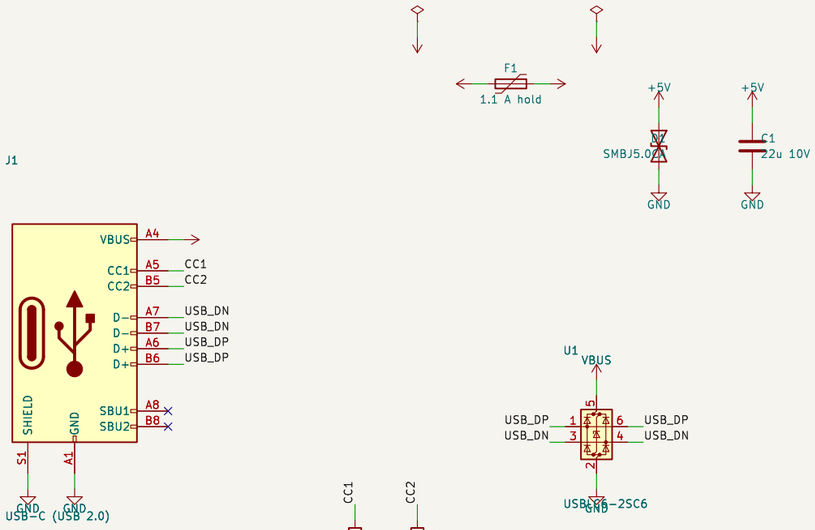
Sheet: /

File: electroNIX–USB.kicad_sch

***Title: electroNIX–USB (Z573M, 4 tubes) – overview***

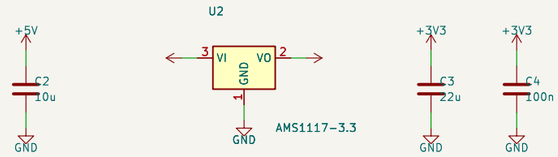
| | | |
|---|---|---|
| Size: A3 | Date: 2026–10–15 | **Rev: 0.1** |
| KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4 | | Id: 1/5 |

## USB-C input (5 V only, sink: 5k1 on CC1/CC2)

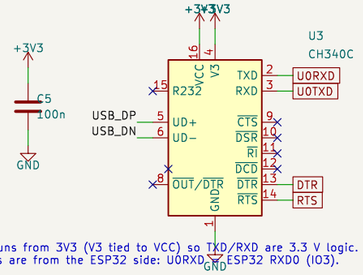



VBUS feeds the boost converter and the 3.3 V LDO only.
Budget from a 500 mA port: ESP32 ~250 mA avg (500 mA WiFi peaks),
HV supply ~0.3 A (4 tubes) / ~0.4 A (6 tubes). Total ~0.5–0.8 A:
use a USB-C charger rated >= 1 A at 5 V (any phone charger), not a PC USB-A port.

## 3.3 V regulator




## USB–UART (programming + serial log)




CH340C runs from 3V3 (V3 tied to VCC) so TXD/RXD are 3.3 V logic.
Net names are from the ESP32 side: U0RXD = ESP32 RXD0 (IO3).

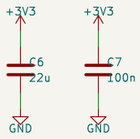
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ESP32 rework of TESTA-QUADRA electroNIXclock 3 (PCB-036), USB powered
**electroNIX-USB (Z573M)**

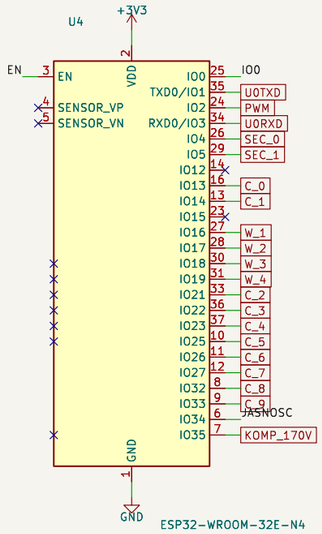
Sheet: /Power & USB/
File: power_usb.kicad_sch

***Title: electroNIX-USB – USB-C power, 3.3 V, USB-UART***

| Size: A3 | Date: 2026-10-15 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/5 |

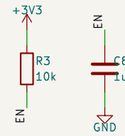
## ESP32 module (GPIO map = stock firmware BOARD_ELECTRONIX_3)



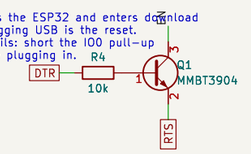




Strapping pins at reset:
IO0 high = run (10k pull-up; BOOT / auto-reset pull it low).
Firmware drives IO0 as its (unfitted) buzzer pin after boot - harmless.
IO2 low/float (boost gate pull-down on HV sheet)
IO12 / IO15 open: internal pull-down / pull-up give the right defaults.
Reserved for anodes W_5 / W_6 on the 6-tube build.
IO5 SEC_1 neon driver base - default state is harmless

## Reset and auto-program (DTR/RTS) - no RESET/BOOT buttons needed



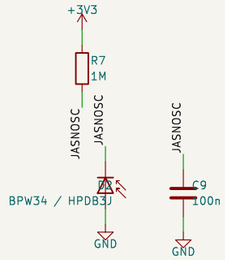

No buttons: the CH340C resets the ESP32 and enters download mode through DTR/RTS; unplugging USB is the reset.
Recovery if auto-reset ever fails: short the IO0 pull-up (the 10k on IO0) to GND while plugging in.




| DTR | RTS | EN | IO0 |
|---|---|---|---|
| 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |

## Peripherals




No buzzer and no RTC: time comes from NTP (firmware BOARD_HAS_RTC 0).
IO0 has a 10k pull-up so it boots normally.

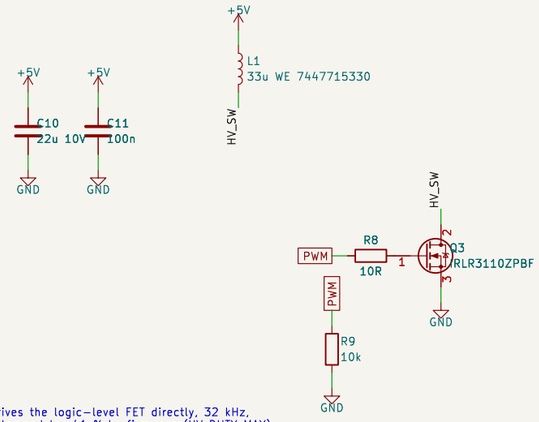
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ESP32 rework of TESTA-QUADRA electroNIXclock 3 (PCB-036), USB powered
**electroNIX-USB (Z573M)**

Sheet: /ESP32/
File: mcu.kicad_sch

***Title: electroNIX-USB - ESP32-WROOM-32E and peripherals***

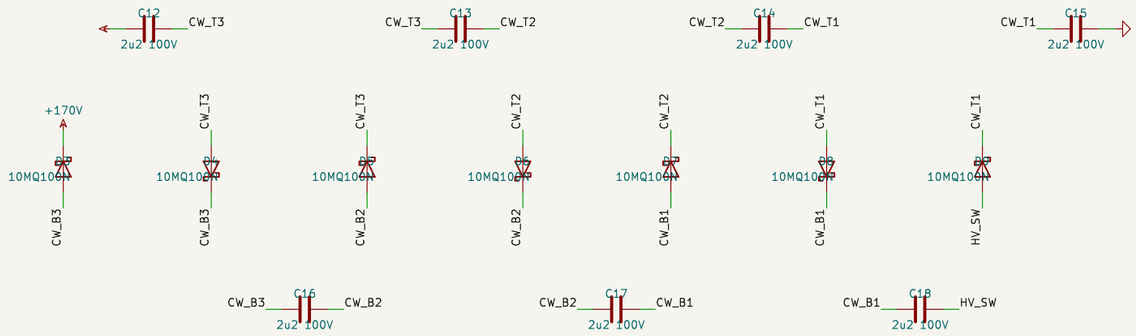
| Size: A3 | Date: 2026-10-15 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/5 |

## Boost switch (as PCB-036)



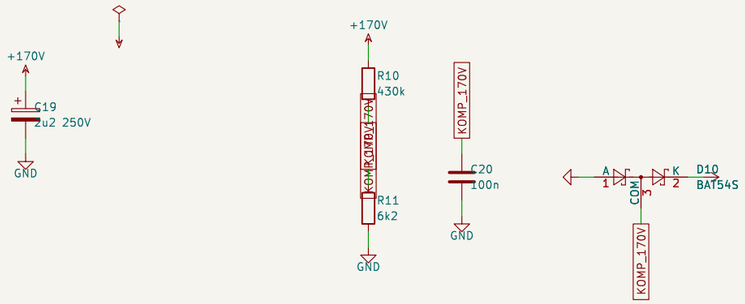

IO2 drives the logic-level FET directly, 32 kHz, duty clamped to ~41 % by firmware (HV_DUTY_MAX). 10k keeps the gate (and IO2) low during boot.

## Cockcroft-Walton multiplier, 7 x 10MQ100N, 7 x 2u2/100V (PCB-036 used 1u)




Each stage adds one boost peak (~45 V at 170 V out); no part sees more than ~100 V.

## Output, feedback (KOMP_170V) and ADC clamp




Divider ratio (430k+6k2)/6k2 = 70.35 = HV_DIV_RATIO in display_testa.cpp.
170 V -> 2.42 V at IO35. BAT54S clamps to 3V3 (the original clamped to 5 V).
Use 1 % parts; 430k in 1206 for working voltage.

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ESP32 rework of TESTA-QUADRA electroNIXclock 3 (PCB-036), USB powered

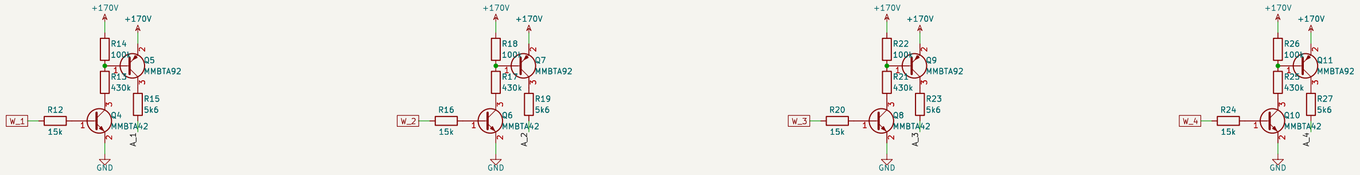
**electroNIX-USB (Z573M)**

Sheet: /HV supply/
File: hv_supply.kicad_sch

***Title: electroNIX-USB – 170 V supply (boost + multiplier)***

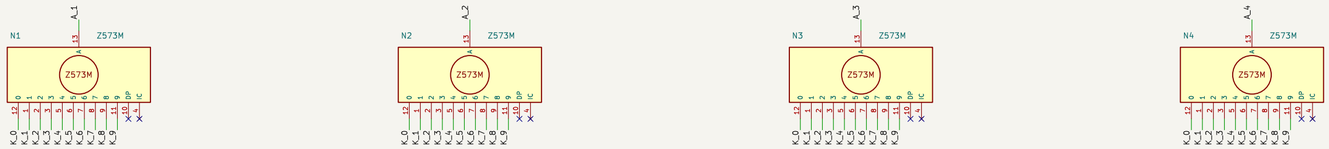
| Size: A3 | Date: 2026-10-15 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/5 |

## Anode switches (one per tube, MMBTA42 level shift + MMBTA92 high side)



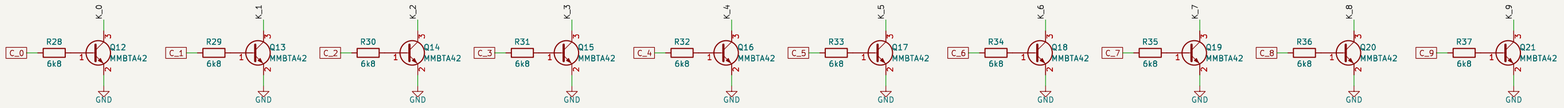

W_n high -> tube n anode at +170V through 5k6: ~5.4 mA peak at 170 V, ~1.3 mA average per tube (Z573M: 1.5–2.5 mA cont., 12 mA pulse).
Peak current follows (V_HV − 140 V): keep the firmware HV setpoint <= 180 V.  6-tube build (--tubes 6): two more of these on IO12/IO15.

## Tubes (RFT Z573M side-view, 13 flying leads on 9.53 mm circle, wire 7 at front)

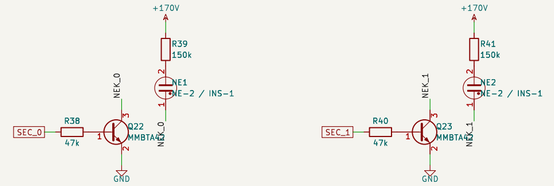



Pin 4 = internal connection (leave open). Pin 10 = decimal point (unused).
Pinout per RFT data book (frank.pocnet.net/sheets/084/z/Z573M.pdf).

## Cathode switches (MMBTA42, shared by all tubes)

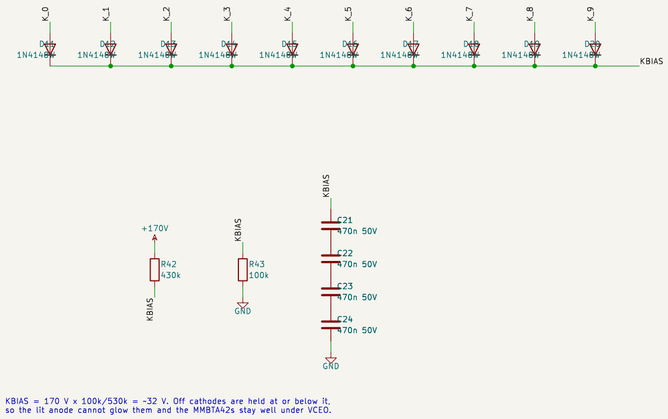



## Colon neons (4 tubes: SEC_0 upper / SEC_1 lower dot.  6 tubes: SEC_0 = HH:MM colon, SEC_1 = MM:SS colon)




Two lamps stacked one above the other in the colon gap, each on its own output.
Firmware drives SEC_0/SEC_1 independently (BOARD_DUAL_NEON). 150k per lamp: ~0.6 mA at 170 V.

## Off-cathode clamp (anti-ghosting, as PCB-036)




KBIAS = 170 V x 100k/530k = ~32 V. Off cathodes are held at or below it,
so the lit anode cannot glow them and the MMBTA42s stay well under VCEO.

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ESP32 rework of TESTA-QUADRA electroNIXclock 3 (PCB-036), USB powered
**electroNIX-USB (Z573M)**

Sheet: /Display/
File: display.kicad_sch

***Title: electroNIX-USB – 4 x Z573M, multiplexed***

| Size: A2 | Date: 2026-10-15 | **Rev: 0.1** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/5 |